

AI Engineer Master Roadmap (16 Weeks)

How to Use This Roadmap

Follow ONLY the resources mentioned for that week. Do not jump ahead. After every concept video, implement it yourself before watching the next one.

Week 1

Padho with Pratyush: Intro, Setup, First LLM Call, System Role.
Krish Naik: LLM, Tokens, Context Window, Temperature.
Build: Basic LLM chatbot.

Week 2

Padho with Pratyush: Prompt Engineering, Pydantic, JSON Output.
Krish Naik: Prompt Engineering concepts.
Build: Job Description Analyzer.

Week 3

Padho with Pratyush: Tool Calling videos.
Krish Naik: What is an AI Agent?
Build: Calculator + Weather + Time tools.

Week 4

Padho with Pratyush: Memory/Agent continuation (if available).
Official OpenAI Docs: Function Calling & Structured Outputs.
Build: Personal Study Assistant.

Week 5

Krish Naik: Embeddings, Vector DB, RAG.
Build semantic search from scratch.

Week 6

Krish Naik: Complete RAG playlist.
Build University Knowledge Assistant v1.

Week 7

LangChain Academy: LangChain basics.
Rebuild your previous project using LangChain.

Week 8

LangGraph Essentials (LangChain Academy).
Krish Naik: LangGraph crash course.
Build graph-based workflow.

Week 9

Official LangGraph docs: Human in the Loop.
Build approval workflow.

Week 10

Krish Naik: MCP Crash Course.
Anthropic MCP docs.
Build local MCP server.

Week 11

Piyush Garg: FastAPI architecture & deployment concepts.
Build backend APIs.

Week 12

Docker + PostgreSQL + deployment.
Deploy project.

Week 13

University AI: Authentication, profile, dashboard.

Week 14

University AI: RAG over university documents.

Week 15

University AI: Multi-agent workflows, reminders, email drafts.

Week 16

University AI: Testing, deployment, README, portfolio polish.

Parallel University AI Timeline

Every Sunday spend 3-4 hours ONLY on the University AI project.

Week 1-2: Project architecture & folder structure.

Week 3-4: Basic chatbot.

Week 5-6: RAG.

Week 7-8: LangGraph.

Week 9-10: MCP.

Week 11-12: Backend.

Week 13-16: Complete production system.

Rules You Must Follow

1. Never watch more than 2 videos/day.
2. Never start a new topic before building the previous one.
3. Push code to GitHub every Sunday.
4. Read official docs only after watching concept videos.
5. Maintain notes with: What? Why? How? Flow? Limitations?